Roy (Yuanxi) Cheng

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Education

University of California, Los Angeles (UCLA) Sep 2023 – Mar 2027

Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering *GPA: 3.75*

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Data Structures & Algorithms, Databases, Machine Learning, Neural Networks & Deep Learning, NLP, Data Analysis & Regression, Data Mining, Probability & Statistical Inference, Monte Carlo, Optimization

Projects and Research

Adrenaline and Cognitive Fatigue Effects on Athletic Performance | *R, ggplot2, tidyverse, dplyr*

- Designed 2^3 factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Verified model assumptions with log transformation and ggplot2 diagnostics, confirming normality and homoscedasticity

Biomedical Tissue Imaging | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article analyzing CNN-based segmentation models and AP metrics to evaluate cytoplasm boundary precision
- Engineered data pipeline to convert 4k+ CytoNuke annotations into COCO to allow training across 3 model architectures
- Improved cytoplasm segmentation by training YOLOv8 in PyTorch, increasing AP50 by 5% over Cyto-RCNN

COVID Pandemic Effects on Secondary Education | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned 100k+ student records by merging multi-level datasets, enabling accurate visualization and analysis in Tableau
- Built time series visualizations by aggregating 4 years of test scores, revealing 15% avg decline in student performance
- Created interactive Tableau map for research website by linking geographic data to score changes across 50 U.S. states

Work and Leadership Experience

Capital One McLean, Virginia

Product Management and Analysis Intern *Jun 2026 – Present*

- Leveraged SQL and Python to analyze workforce data, identifying factors tied to successful organizational transitions
- Engineered a 6-stage Python pipeline through anonymization, BGE filtering, and FAISS retrieval for LLM DWA scoring
- Built a custom LLM client with sliding-window rate limiting and retry-aware backoff for reliable inference throughput

Data Science Union Los Angeles, California

External Vice President *Dec 2025 – Present*

- Directed client partnerships, marketing, and professional development initiatives to expand opportunities across members
- Managed 3 directors and coordinated executive priorities, ensuring seamless execution of quarterly and annual objectives
- Cultivated industry and alumni partnerships to secure client projects, increase visibility, and expand career programming

Kohl's Los Angeles, California

Project Manager and Data Analyst *Jun 2025 – Mar 2026*

- Analyzed 8.5M+ logistics records using SQL and Python, resolving inconsistencies and validating supply chain data
- Conducted EDA in Vertex AI Workbench, engineering predictive features to enhance shipment transit time forecasting
- Developed and deployed two-stage XGBoost forecasting model, improving shipment arrival prediction MAE by 83%

ACTUM Digital Prague, Czech Republic

Data Analyst and AI Intern *Jun 2025 – Aug 2025*

- Managed project workflows in Jira/Confluence, presenting structured weekly updates that improved delivery timeline
- Engineered Python scripts with agentic methods to filter files, generate metadata, and embed into Milvus vector database
- Authored prompting and integrated MCP server to create AI assistant, co-presenting live demo to ACTUM executives

Million Dollar Baby Los Angeles, California

Data Analysis Consultant *Apr 2025 – Jun 2025*

- Analyzed $10M+ ad spend data with SQL to extract insights that informed growth strategy and investment decisions
- Presented CAC and RoAS assessments across attribution windows to guide stakeholders on marketing spend allocation
- Built a dashboard of 20+ marketing KPIs in Excel to highlight performance trends to brand managers and stakeholders

Technical Skills

Languages: Python, C++, Java, SQL, R, JavaScript
Libraries/Frameworks: PyTorch, pandas, NumPy, scikit-learn, TensorFlow, Keras, XGBoost, Matplotlib, Seaborn, ggplot2
Tools & Platforms: Git, GitHub, GCP, BigQuery, Snowflake, Vertex AI, Tableau, Jupyter Notebooks